Maja Lind, Matilda Jansson, Viktor Sersik, Kassie Ka Chai Kuek

2024

WORKBOOK

The making of a shared kitchen booking system

PART 1: DISCOVER

The start of the project: Defining a user group and discovering the problems

→ The design brief

When we received the design brief, our primary focus was on the concept of people interacting with one another. Interdependence, by definition, means:

"Interdependence refers to a mutual reliance between two or more parties, systems, or entities, where each one depends on the others to function effectively."

We began by imagining scenarios where people might need assistance and how others could step in to provide it. Additionally, we considered situations where people are naturally required to interact, such as ordering at a fine dining restaurant. For us, these examples illustrate interdependence, moments where interaction is both necessary and mutually beneficial. Interdependence is about shared experiences and about understanding how individuals rely on one another in those shared experiences. Therefore our interpretation of the design brief was to design something that supports people in their shared experience. In other words, something that highlights coexistence, how people

depend on each other, their shared experiences and how they maintain them. Importantly, these shared experiences can occur between individuals living together.

→ User group

In order to decide on a specific user group we started with a mind map, thinking about different activities. This is what we came up with →

We had a lot of discussions and ideas and eventually we decided on: *People living in a corridor.*

However, that is a bit too broad, so after more analyzing and discussing we landed on the final user group: *Students in Stockholm living in a student corridor*

→ Collecting data

The next step was to start collecting data. Two methods were used to do so:

1. Secondary data analysis

The data collection was carried out by researching the website *Stockholms Studentbostäder* (sssb.se). We looked for prices, sizes of rooms, how many kitchens there were in a corridor, queue days needed and what supplies the kitchens had.

2. Interviews - 6 interviews were done!

We conducted semi-structured interviews with six different people. All who have either lived in a corridor in the last 5 years, or are currently living in a corridor.

A typical corridor room in Stockholm with rent, size and queue days

Questions for a Semi-Structured Interview

1. Have you lived/Do you live in a student corridor?
2. For how long?
3. Where?
4. How many people live in your corridor?
5. What areas do you share with other people?
6. What made you choose to live in a corridor?
7. What is your general opinion?
    a. Cohesion
    b. Health
    c. Cleanliness
    d. Community
    e. Cooking
    f. The kitchen
    g. Security
8. What do you think could improve…
    a. Cohesion
    b. Health
    c. Cleanliness
    d. Community
    e. Cooking
    f. The kitchen
    g. Security
9. How well did you enjoy living there, on a scale of 1-10?
    a. What would increase the score? / What would make it better?
10. How long could you see yourself living in a corridor?
11. What was the best thing about living in a corridor?
12. What was the worst thing about living in a corridor?

Interview questions

Important highlights

The most important parts we took away from this phase were the different ways of interpreting the brief. It is very important to have a clear interpretation before starting the project. Furthermore we had some trouble in the beginning narrowing down our target user group. It started broad and then we had to narrow it down even more. Something we feel like we managed to do! It is also important to note that this is the first step in the double diamond, getting a lot of information and thoughts, before we narrow it down the first time.

Maja conducting an intereview

PART 2: DEFINE

The making of the problem: Analyzing the data collection and defining a problem

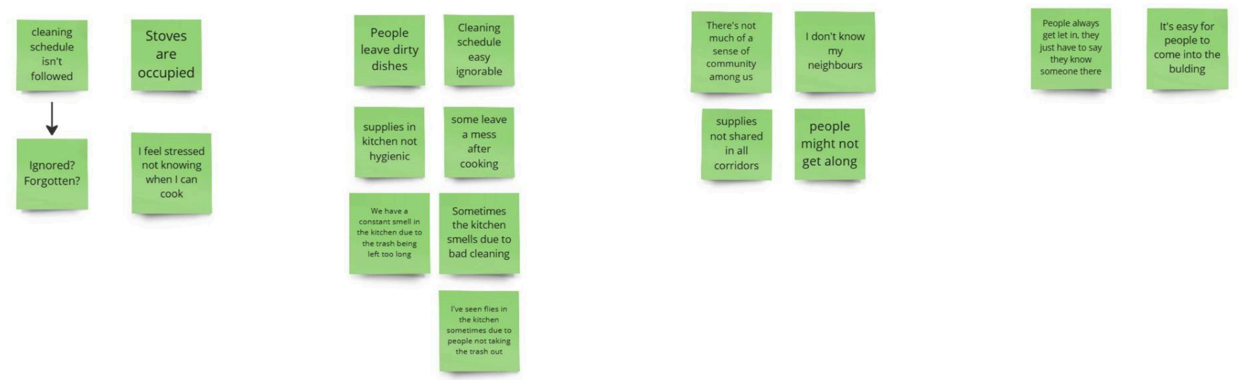
→ Analyzing the data

The next step in our project was to analyze the data we had collected, and define a problem. This is the next step in the double diamond method where we narrow it down.

1. Affinity Diagram

The data analysis started with an affinity diagram to get an overall view. We listed things found on the website as well as things said in the interviews. Afterwards we group them into clusters based on common themes. After that we named the clusters, categorizing them into their main themes.

Problems found during the discover phase



Problems sorted into clusters

Reviewing and prioritizing

2. Personas & Scenario

After the affinity diagram, personas and a scenario were created to help understand the possible problems even more. The personas were made to reflect the four main topics found in the affinity diagram. We chose to have four personas in order to make sure we covered all the main themes and have more personas to work with in the scenario.

Scenario 1: Everyone's in the kitchen, wanting to use it at the same time

The corridor kitchen has two stoves combined with ovens, and a microwave. There is a shared dining table with a few chairs around it. It's 19:00, and Tai Man walks into the kitchen to cook something simple for dinner. He looks through his cabinet, and decides on instant noodles. While filling up the common pot with water, Max enters the kitchen. He grabs a frozen pizza from the freezer, as he does every day, and pre-heats the oven. He takes a seat in one of the dining chairs. Cardi walks in with three friends, and announces that they will have a hang out in the kitchen. She hasn't asked the others about it. They all go to the other available stove and start frying some nuggets. The kitchen is now crowded.

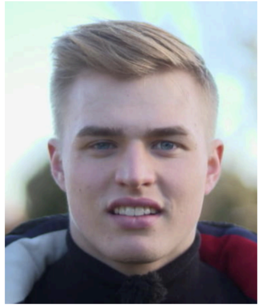
Lena walks in, a bit late to her football practice. She wants to cook sweet potatoes in the oven, but the only one available is the one where Tai Man is standing cooking noodles. She asks, a bit stressed, if she can use the oven. Tai Man doesn't answer. Lena thinks that she can fry the potatoes on the stove instead, but sees that both stoves are occupied. Lena does NOT want to eat microwave food, since she thinks that is too unhealthy.

She starts to freak out, and when no one wants to give her a place to cook, she leaves the kitchen, stressed. Cardi and her friends laugh a bit, Tai Man looks uncomfortably down on the floor, and Max has his headphones in, so he didn't even notice.

Person 3

Name: Lena Andersson
Age: 24
Gender: Female
Origin: Luleå, Sweden 🇸🇪
Profession: Student BSc "Vehicle engineering" at KTH, 3rd r
Private life: Single, family living in Argentina
Likes: Playing football and cooking healthy food
Needs Wants & Expectations: Wants to work as a space engineer at NASA after her degree
Motivations & attitudes: Thinks school is alright, not the most fun but is has to be done in order to get a degree and work at NASA
Frustrations: Not having a structured schedule
Relevant Tasks: Is a part of a football team and practises 3 times per week
Channel or product preference: Youtube

Person 1



Name: Max Mustermann

Age: 18

Gender: Male

Origin: Salzburg, Austria 🇦🇹

Profession: Student BSc "Information and Communication Technology" at KTH, 1st year

Private life: single; plays amateur football, but will tell you that if he did not tear his ACL he would be a professional footballer; goes out every friday and saturday, gets way too drunk, and talks to everyone

Needs Wants & Expectations: parties, drinking, fun

Motivations & attitudes: does his study because his parents wanted him to, moved to sweden because of the beautiful girls

Frustrations: people being too stuck up; people being annoyed easily; others not understanding his style of humour

Relevant Tasks: Likes playing football with friends, drinking beers afterwards, and going to the club later

Channel or product preference: Always does the bare minimum, i.e. eats frozen pizzas 4 times a week and tries to copy all his homework

Person 2

Name: Belcalis "Cardi" Marlenis Almánzar

Age: 21

Origin: New York, USA 🇺🇸

Profession: Exchange music student at SU

Likes: Music, friends

Dislikes: Cooking, cleaning

Private life: Single, mostly hangs with friends and does not spend much time in the corridor. Would like to have a house to have social gatherings in, but does not feel like the corridor is a suitable place.

Needs, Wants & Expectations: Fame, prefers to have things done for her

Motivations & attitudes: Studies to become rich, doesn't want to live in a corridor

Frustrations: Getting told what to do

Relevant Tasks: Likes throwing and attending parties

Channel or product preference: Likes apps that helps her avoiding chores, like UberEats

Person 4

Name: Chan, Tai Man

Age: 25

Gender: Male

Origin: Hong Kong 🇭🇰

Profession: International KTH MSc Student

Private life: His girlfriend lives in Hong Kong and they are in a shaky long distance relationship

Needs Wants & Expectations: needs food, water and good grades.

Motivations & attitudes: Hopes to finish his studies as quick as possible so he can go home, does not like going out and stays inside all day

Frustrations: Doesn't know how to cook but delivery foods costs has to cook anyway, cannot talk to the opposite gender normally

Relevant Tasks: Work**,** study, cooking, shower, talking on the phone with his girlfriend

Channel or product preference: anything not heavy and easy to operate

→ Defining a problem

During our analysis we saw a big trend. We understood that the kitchen seemed to be the constant problem for all the participants. That together with the literature review that revealed that there are often 9-14 people sharing a kitchen, made us decide that this was the problem we should look into. However, “the kitchen” is a bit too broad of a problem. Looking further into it, we saw that there were two parts of the kitchen that were especially difficult: *keeping it clean, and having enough space for everyone.* That is, the kitchen being occupied. People felt it was hard to plan their day not knowing when the kitchen is free due to the lack of space. After identifying a problem, we had to narrow it down. In order to specify these trends and make the problems more specific and clearer, we created POVs and HMW-statements.

1. POV

We created a bunch of POV-Statements which we later cut down to only one specific statement in order to have a very clear idea on what problem we wanted to focus on!

1. Lena needs to cook dinner because she has football practise
2. Lena needs to arrive at a clean kitchen at 17.00 because she has practice later and needs to eat
3. Lena needs to know when the kitchen is free because she has to plan her day
4. Lena needs to plan her day because she has a lot of schoolwork to do as well as practice in the evening
5. Lena needs to eat healthy because she is playing football on a high level
6. Max needs to cook some food at 04:30 because he just got home from a night out
7. Max needs help with his study because did not do anything for the entire period until week 7
8. Max needs store alcohol in his fridge because he just likes to drink

Kitchen needs to be free and clean in order for people to be able to use it!!

2. <u>HMW</u>

We also created HMW-Statements. We didn't create as many but had once again a hard time cutting formulating only one very specific statement.

1. How might we make it easier for people to plan their day
2. How might we help people not feeling stressed about cooking
3. How might we make it easier for people to share a kitchen
4. How might we make sure people clean after themselves

We saw strong connections between cleaning and spacing, and had a hard time in the beginning separating them. However, through our personas and HMW-statements we could see that the space issue was the main thing. Sharing a kitchen is about sharing the space, and even though it may not be nice that it is dirty, people will still be able to cook there. However, if the kitchen is always occupied, they won't. Therefore we choose to focus on that, and try narrowing that problem down as much as possible. We finally landed in the POV and HMW:

Lena needs to know when the kitchen is free because she has to plan her day

How might we make it easier for people to plan their day

and decided to work based on that!

<u>Important highlights</u>

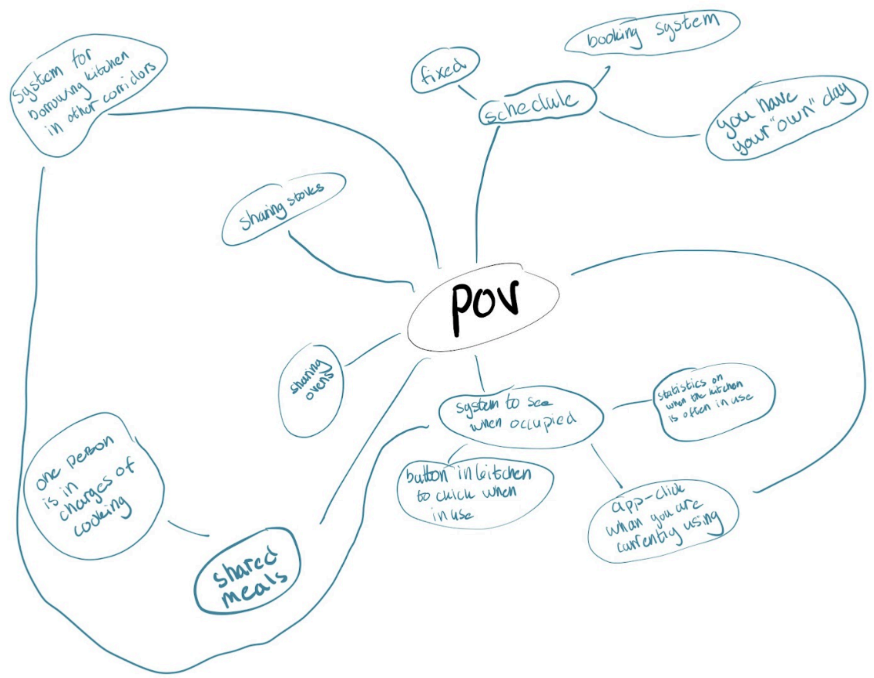
The most important part to highlight in this stage of our project is the importance of having a very clear idea of the problem, and being as specific as possible. As mentioned we had some issues deciding on only one problem as well as narrowing down that problem as much as possible. However, we eventually landed in a spacing issue, with our POV "*Lena needs to know when the kitchen is free because she has to plan her day*" and HMW "*How might we make it easier for people to plan their day*" as foundations. We felt these connected in a good way to there not being enough space for everyone in the kitchen. Since one issue that leads to there not being enough space for everyone in the kitchen is because everyone is cooking at the same time. Seeing that issue from a different angel shows us a scheduling issue. The kitchen needs to be shared between people in a good way, for example not everyone can cook at the same time, they need to share the time. With that comes knowing when other people are cooking and for that to be a possibility, people need to be able to plan their days and lives.

PART 3: DEVELOP

The solutions to the problems: Using ideation methods and the making of low-fi and hi-fi prototypes

→ Ideation

The next step in our project was the development phase. This is the phase where we start coming up with ideas and solutions to our specific problem. In the double diamond, this is where we widen the diamond again, the third phase. This started with ideation and three methods of ideation were used.



1. Brainstorm

In order to get as many ideas as possible we started out by having a brainstorming session. We thought about all the different aspects to sharing a kitchen, as well as ideas that could help!

2. Worst possible idea

We also did the worst possible idea method. This helped us to think outside of the box and see the problem from a different perspective.

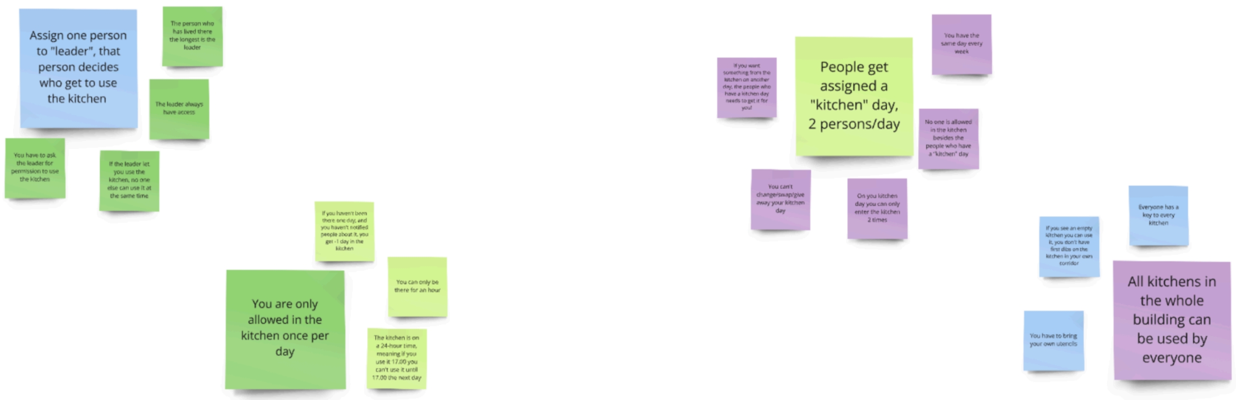
In this method we also got the chance to “work backwards” and see what we don’t want as well as see what kind of features we want to avoid. In addition to that it helped us create more creative solutions to the problem. This is when we realized we wanted to create an app, since a lot of our other ideas contained traits we do not want.

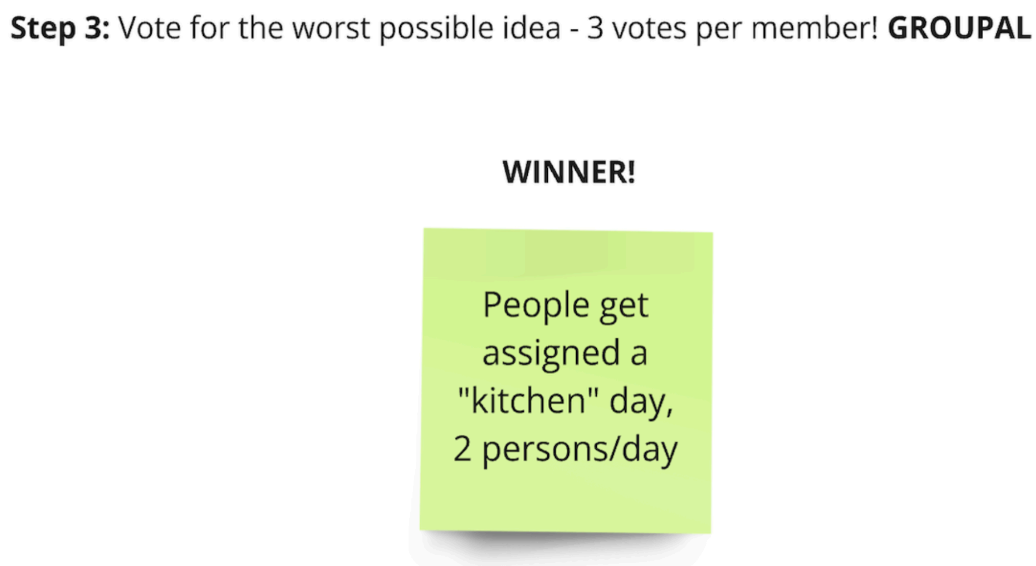
Step 1: What are the worst possible ideas you can think of? **INDIVIDUAL**

We started by individually writing down the worst possible ideas we could think of by making notes on the miro board. We spent 5 minutes on this.

Step 2: Give your worst idea to another person. Their job is to extend the idea in another round of "worst possible idea". **INDIVIDUAL**

After that we chose one idea to give to another person who had to extend it and think one step further by also writing down points in miro. On this we spent 15 minutes.



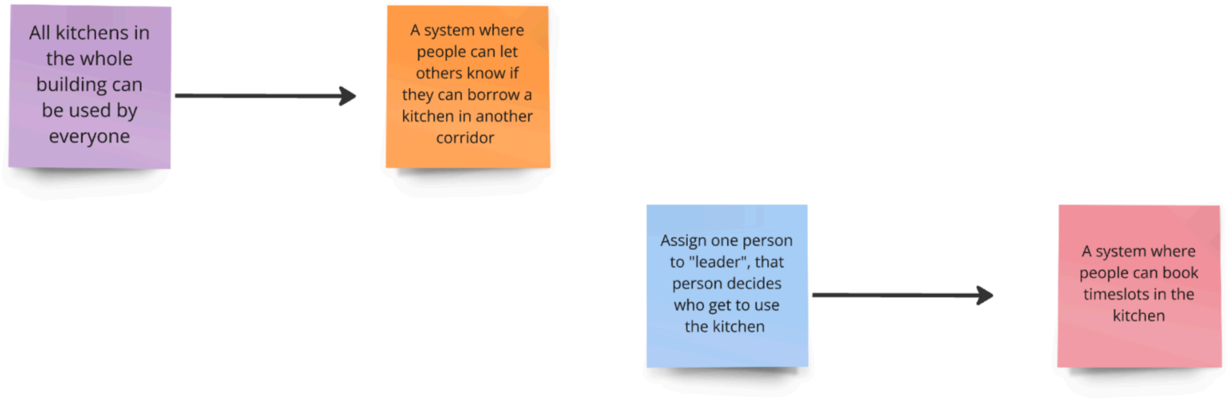


After that we presented the ideas we had come up with for each other and voted on which one was the worst. Everyone got three votes each. For this we set 30 minutes.

When we had our winner, we came up with common attributes of that idea. This was done as a group. For this we set 20 minutes.

Step 5: Can we turn the bad ideas into good ones?

Lastly we tried turning these bad ideas into good ones, again this was done as a group and we spent about 20 minutes.



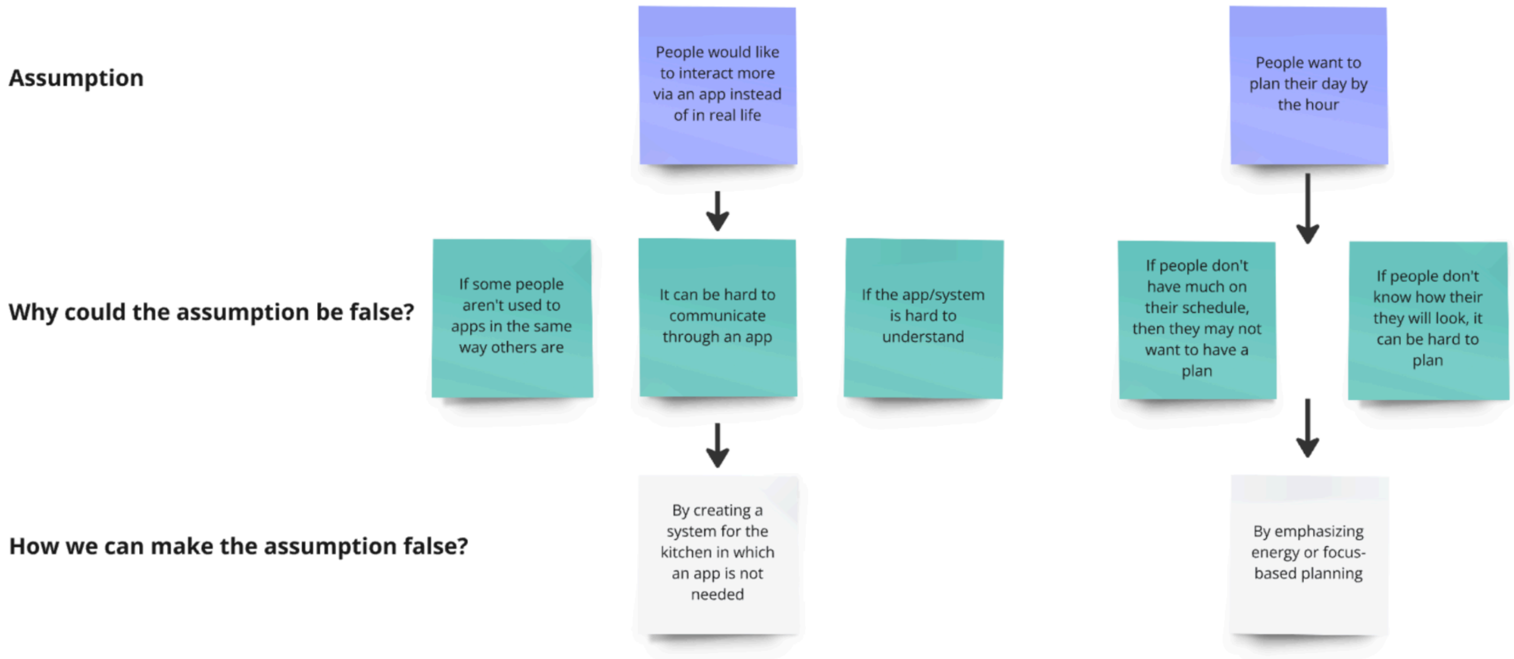
During this method everything was written in Miro and timed by an alarm on the phone. The notes are organized by color with one person always using one color.

One member made sure that once the alarm sounded, the group moved on to the next one. Repeating what the next step was and making sure everyone understands. Acting as a moderator. In total this method took about 2h.

3. Challenging assumptions

The last method we used was the “Challenging assumptions” method. This was very helpful in discovering new possibilities by questioning the "rules" and a good way to “double check” ourselves and make sure we don’t get stuck in our own ideas and assumptions. By posing the questions *“Why could the assumption be false?”* and *“How can we make the assumption false?”* we had to go against ourselves and challenge ourselves. The first step was to list out assumptions. We then thought about how these assumptions could be false. Lastly we thought about how we could make the assumptions false.

This method really helped us think outside of the box and made sure we questioned our ideas. The main insight was that creating something for a lot of people isn't always easy. Even though an app might be perfect for someone, it does not fit everyone. Furthermore, not everyone workes the same so not everyone wants to plan their day. However, we did end up sticking to our assumptions in the end. This is because we felt like an app was the best for our problem, due to the results of the worst possible idea method, but it still worked as a good food for thought.



Summary of ideation

Through our different methods of ideation we came up with a lot of ideas. Brainstorming helped us to really think about as many ideas as possible and branch out in all kinds of directions. Worst possible idea helped us create more creative solutions and gave more insight on what we want and don't want. It was through this method that we decided that we wanted to do an app since a lot of our other ideas contained traits that we do not want but an app seemed to work well for this problem.

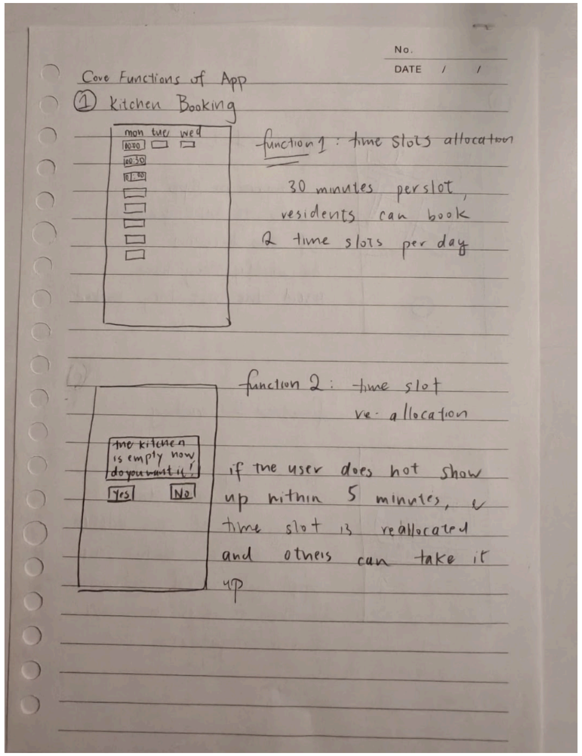
The challenging assumptions method made us go against our idea of an app and made sure we didn't get stuck in our own assumptions. Even though we have decided to stick with the app, it served well as a food for thought making sure we question ourselves to give the best solution as possible and not stick with the first one we came up with!

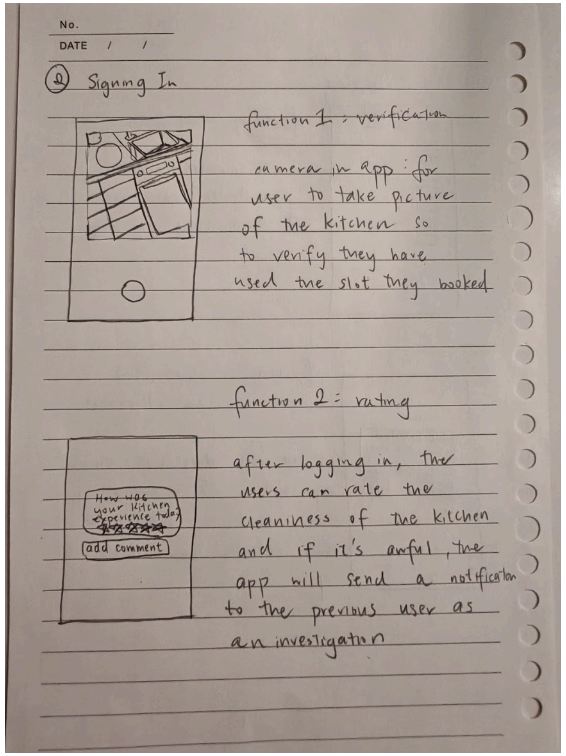
All in all we finally landed on the idea of a booking-app where people can book time slots to be in the kitchen. One of the main problems we realized was that it was hard to know when you could cook, and that most people usually went into the kitchen at the same time, making it impossible for everyone to be there. Therefore, a booking system where people can book time slots felt like a good solution. Not only does it help people to plan their day, by knowing when you will have dinner and not having to check every once in a while if there are any spots available, it also makes sure that everyone gets time in the kitchen thus giving space to everyone and sharing it in a good way.

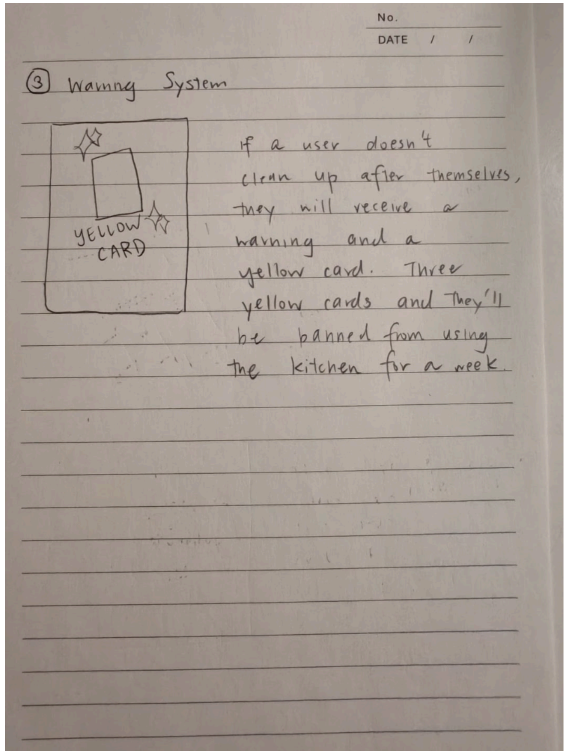
Final decision: *A booking app for the kitchen*

→ Low-Fi prototype

Once we had our idea we needed to make a low-fi prototype in order to decide on the basic functionalities of our app. Here are some pictures of that:







→ Evaluation of Low-Fi prototype

Once we had done the low-fi prototype we tested it with some users. We let them look around on the papers and explained our idea to them, then they got to ask questions.

Overall we got good feedback and people understood the meaning of the app. We got some feedback on how the yellow cards will work, that is, how will you know who left the kitchen messy. We also got the question if you book the whole kitchen when you book a timeslot, or if it is only a part of the kitchen. We took this feedback into consideration and incorporated it in the hi-fi prototype.

Important highlights

In this phase of the project, it is important to highlight that it is crucial to start off with a lot of ideas, in order to narrow them down in the end. We did this by using 3 ideation methods. It is also important in this phase as well to be as specific as possible, don't think you can solve all problems at once. We had a hard time choosing between the problems of cleanliness and space in the kitchen, however we did end up narrowing it down to only the space. After we focused on that part, it gave us the possibility to incorporate a bit of the cleanliness as well, which we think made a good fit and a helpful addition to our app.

We want to really highlight the narrowing down part, which was really important to get a more specific idea to start with! We also want to highlight the prototyping part. We briefly started out with a low-fi prototype in figma which we quickly got feedback on from our supervisor who thought it was too detailed. We got the recommendation to start over which we followed since the idea of the low-fi prototype is to focus on the core functionalities we had come up with already, and we noticed we quickly got carried away when using Figma. Even though we had our core functionalities, when prototyping one of them we quickly got into the detailed part when doing Figma since we enjoy doing that. Fortunately we did not get far, just briefly getting started, before getting the reality check from our supervisor. Therefore we realized the issue and started over and started out with doing a paper prototype only focusing on the core funcitonalltíty of our app. This was really helpful when designing the hi-fi prototype since it helped us put our idea onto paper first, and we are thankful we started over and not got too carried away!

Too detailed lofi prototype

PART 4: DELIVER

The Hi-Fi prototype

→ Hi-Fi prototype

For our hi-fi prototype we decided to work in Figma. Since we are making an app, we thought Figma is a great tool to design and display our ideas. From the feedback we got on the low-fi prototype, we decided to incorporate a camera function for reporting a messy kitchen, where the app keeps track of who was in the kitchen last and sends the warning to them. Furthermore, we made sure that it is possible to only book a part of the kitchen, in order to make sure more people can use the kitchen at once, making it easier to share.

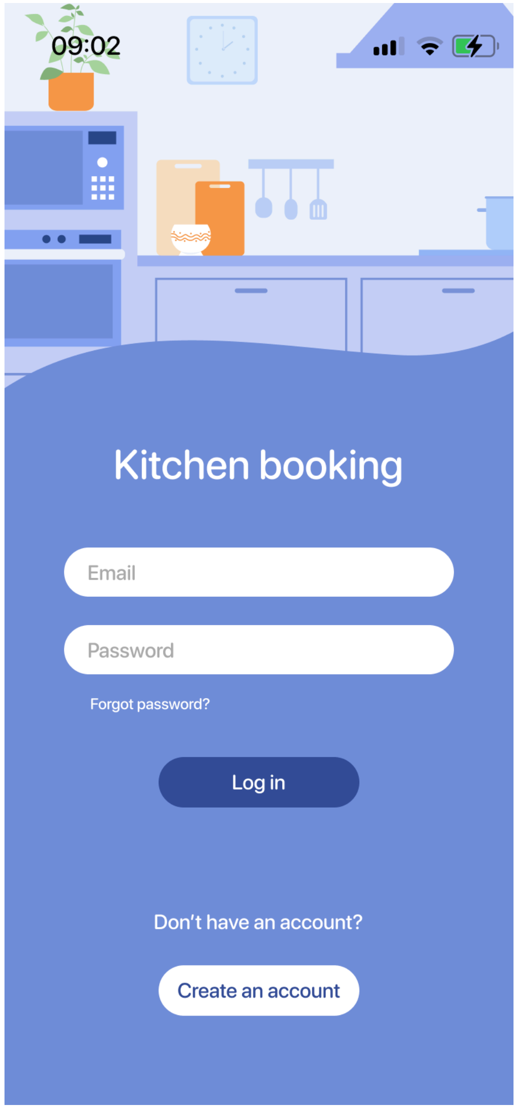
Functionality

Our app is a booking system for booking stoves and ovens in the kitchen. It is made for people living in a student corridor, where the kitchen is shared between multiple persons.

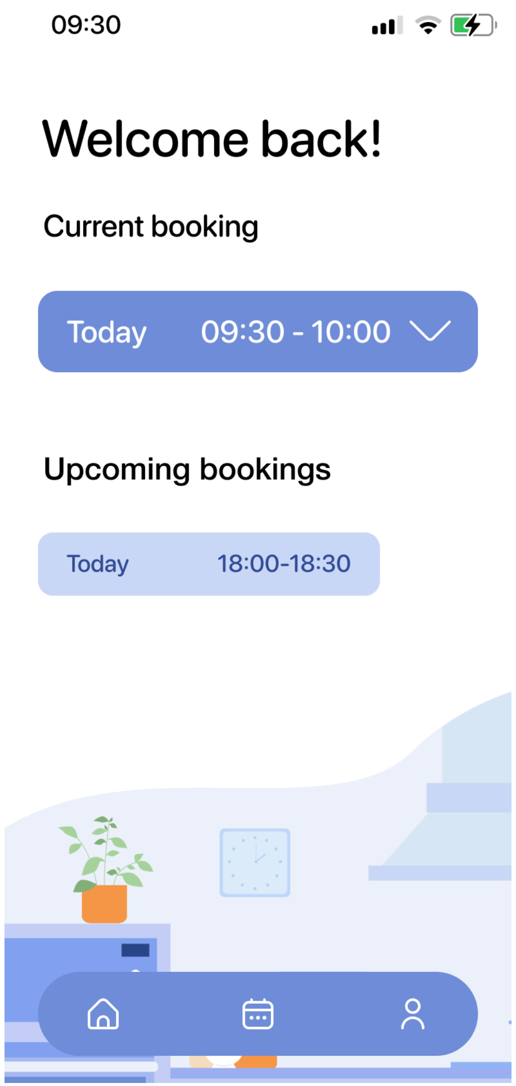
It is made to make it easier for people to plan their day, having a set meal time, and making it easier for them to share the kitchen. For many people, the stress of not knowing when you will be able to cook is tiring. Having to constantly check when it is free, and hurrying when it is empty makes other daily tasks impossible to plan. Our app helps with that. You will be able to book a slot ahead of time, making sure you will know in advance when you will be able to have a meal.

Furthermore, as we mentioned before, we had a hard time deciding between the space of the kitchen and the cleanliness. We wanted to focus on only one problem in the beginning, to make it as specific as possible. However, since we felt like they are connected, if the kitchen is messy you will have to clean it before you can start cooking, we saw it necessary to incorporate a bit of cleanliness as well in the end.

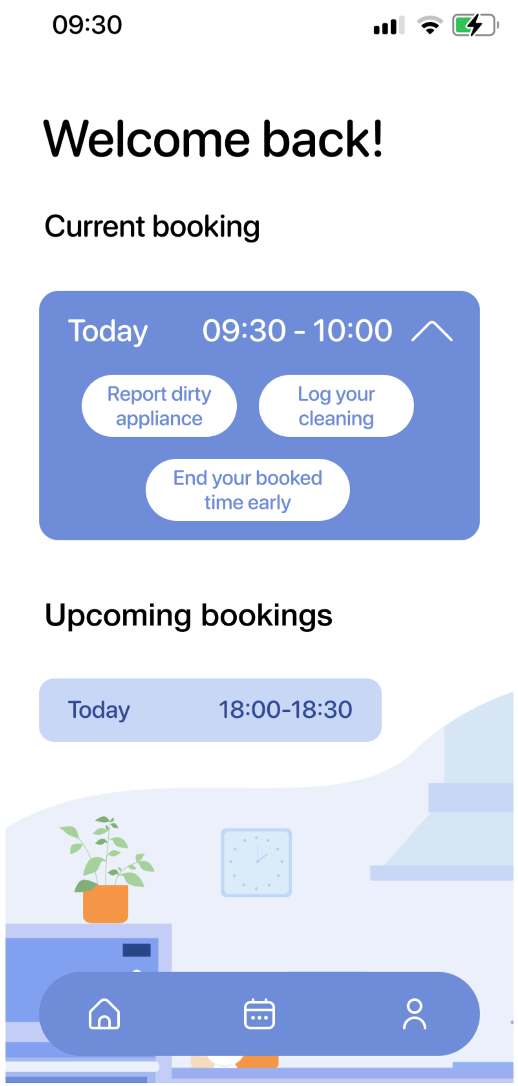
Therefore we have set up the “yellow card” system, where if you don’t clean up after yourself you will get a yellow card. If you get 5 yellow cards you won’t be able to book a new kitchen spot.



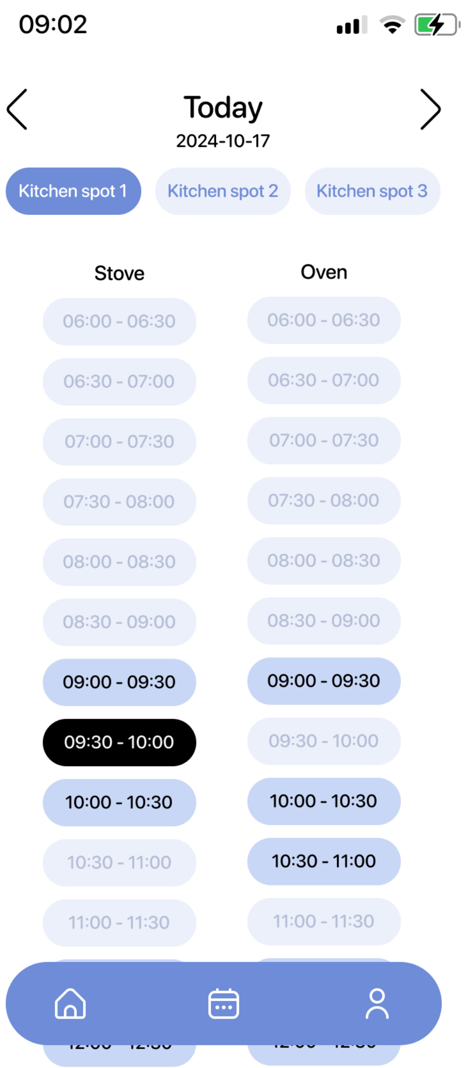
Login page



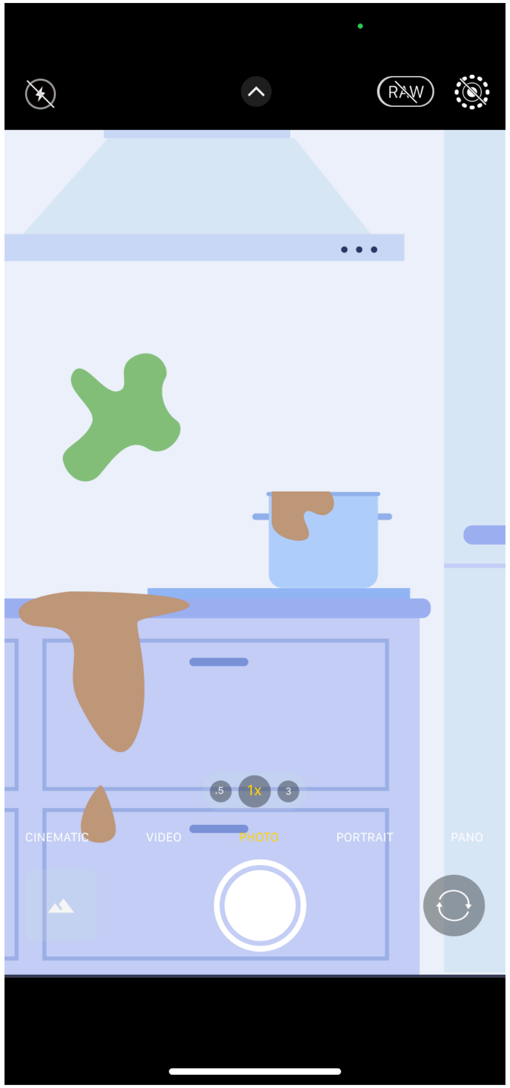
Home page



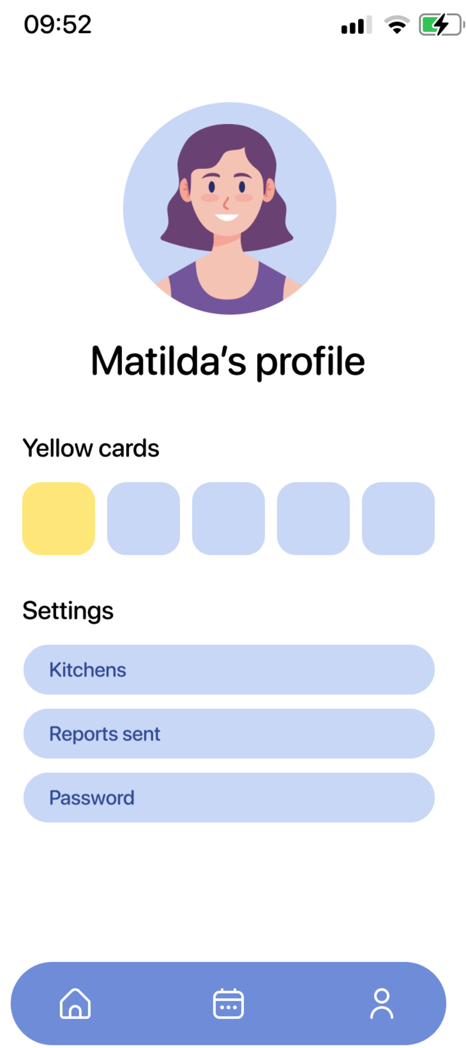
Extended card with options for current booking



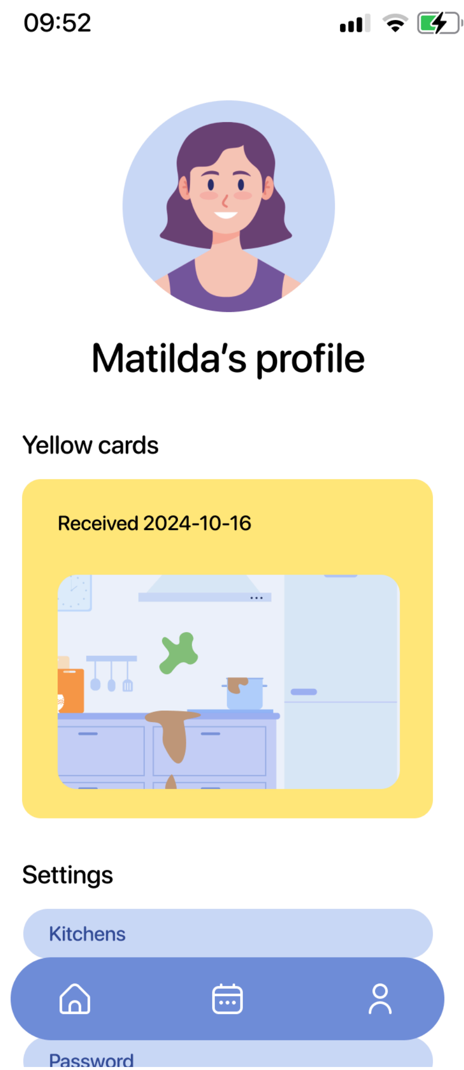
Calendar with a booked time slot



Using camera to report a mess



Profile page



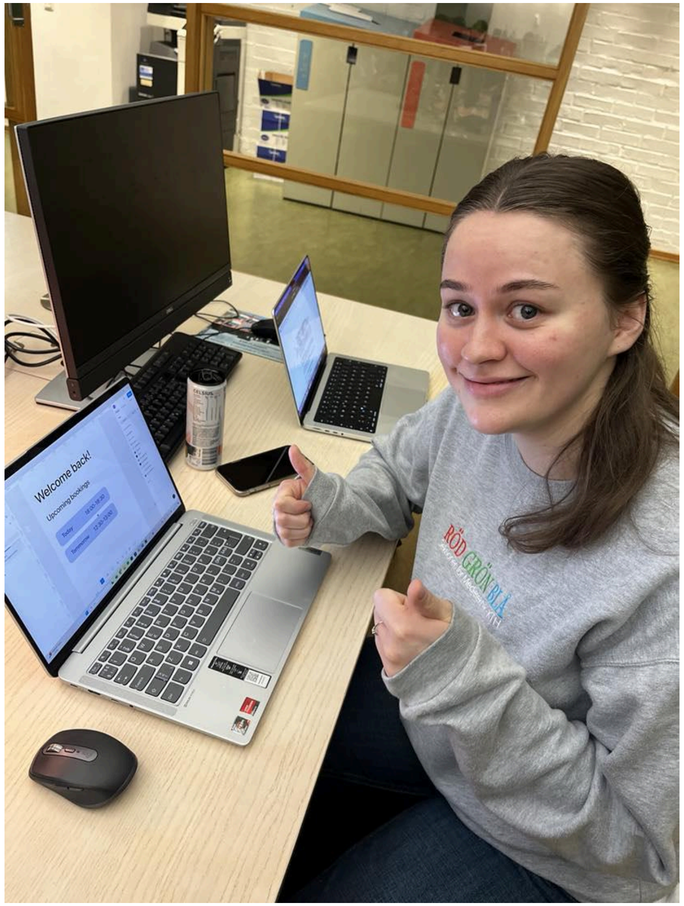
Expanded card to show someone else's report sent to the user

Design

Our app utilizes the whole kitchen. Usually people don’t need both the stove and the oven when cooking, our app makes it possible to book both or either one of them. Giving access to more people to use the kitchen at a time. This is a great way of sharing the kitchen and making sure it is utilized to its best capacity. Furthermore a time slot is 30 minutes long, giving room for both smaller meals without taking up unnecessary time, as well as bigger meals by booking more spots.

The design of the yellow cards allow others to report someone if they haven’t followed the rules of cleaning up after themselves. With the system of having 5 yellow cards getting you banned from booking, it will motivate people to also clean after themselves, hopefully reducing the problem of having dirty kitchens.

Simplicity was the main design concept guiding our decisions when making the figma prototype. Our goal was that the user shouldn’t have to know anything in advance to be able to use the app. To add to the simplicity, we worked a lot with color and symbols instead of words. This was hard finding a balance for, something we got confirmed during the user tests. We also wanted to have as few functionalities as possible to not confuse the user, but this also turned out to be more difficult than we thought, and maybe it backfired a bit.



Matilda prototyping

Evaluating the product: Future improvements

→ Evaluation of Hi-Fi prototype

Once we were done with our hi-fi prototype it was time to test it. In total, five user tests were made, and these are the results:

Thinking out loud method

In order to test our prototype we used the *think aloud method* in order to get a better insight in the users mind and thoughts when they used the app. We set up a series of tasks for the users to do and had them speak through the process. This way we could notice straight away if something was unclear or if something was missing. After each task, we also made sure to ask if anything was unclear, hard to find etc., just in case the user didn't mention it during the task.

Instructions

These are the instructions given to the users and the tasks they were asked to do. The tasks were designed to include every feature of the app, to test every interaction possible in the app.

The outcome

In our evaluations we got a lot of helpful feedback. The main points were the following:

Instructions

This is an app for booking stoves and ovens in student corridor kitchens. For this test, I will ask you to do some tasks as a student living in a corridor. Let's start!

1. It's 9 in the morning. You feel like cooking oatmeal for breakfast. Book a stove from 9:30 - 10:00.
2. Your booked time has arrived. When you arrive to your stove, it's very messy. Report this mess in the app.
3. You have cooked your oatmeal, and cleaned up. Log your cleaning to prove that you didn't leave the stove messy.
4. You still have time left of your booking, but you feel done. End your booking time early so that someone else can book the rest of it.
5. You're planning ahead, and want to make sure you can make lunch on your lunch break tomorrow. You're making oven roasted potatoes. Book an oven tomorrow at 12:30 - 13:00.
6. Check that your new booking can be found under upcoming bookings.
7. Someone else thought you didn't clean up properly after yourself when you cooked dinner yesterday. They reported the dirty appliance, and you received a yellow card. Check the yellow card.

1. Not part of the tasks, but there is no way to cancel a booked kitchen spot.
2. The stove and oven heading could stick when scrolling in the calendar, so that you still see it and don't forget which one is which.
3. A header on the homepage saying "no current bookings" would be nice, so that you don't get confused when it suddenly doesn't exist anymore (since there is a "current booking" header during a booked time.
4. Unclear that the yellow card could be clicked when not expanded, so something would be needed to indicate that.
5. People tried clicking the cards under "upcoming bookings" as they thought they could get more info about their booking, such as which stove or oven they booked.
6. The user wished some more info was provided about what happens after five yellow cards. That could be clearer under that heading.

All in all it was a very successful evaluation, with a lot of very good feedback! The users understood the main concept of the app and were able to carry out the tasks. However, we got a few inputs on the design and clarity of some of the features, which we could incorporate in the next version of the app!

Overall, the test users liked the idea of the app, and liked the overall design and features. No one needed guidance during the test to be able to finish a task, and the points of improvement focused more on clarity. Although the feedback above is for improvement, we also got a lot of good feedback about the simple color scheme, the clear elements, and the simple but useful features.

→ Future design improvements

For future design we should first implement a function to make sure people are able to cancel their spot in the kitchen as well in order to make sure a kitchen spot does not go to waste. We should also work a bit on the design for more usability. First off, make sure it is possible to click the "upcoming bookings" in order to get more info, for example if the oven or the stove is booked. It was also not clear that the yellow card could be clicked on, we should add design to make that clearer. Furthermore, we got a comment that having a "No current bookings" would be nice, so that is also something to implement. It would also be nice to make sure the "stove and oven heading" would stick when scrolling in order to still see which appliance you are checking the time for. This is a valuable function in case you forget which side is which when scrolling.

Besides improving the design from the user feedback, we also have some points regarding the aesthetics of the prototype that we feel like should be improved. As for now, all the elements are a bit too big for a phone screen. This is because the prototype was designed on a computer screen, and we found it difficult to find the right sizing. We realized too late that the figma prototype can be viewed on your phone and changed in real time, which makes designing for phone screens so much easier. This is definitely something we will keep in mind for later.

Important highlights

This is the part of the process where the product comes to life. It is important to remember that we who have worked with this project might have a clear idea of what we want it to be, however, it might not be as clear to other people. Therefore it is extremely necessary to do usability testing and a lot of evaluation, to make sure that the design is working the way we want it to be, and to be able to create the best product possible. This is an important step in the process that can not be forgotten! We want to highlight all the good feedback we got and we enjoyed reflecting on it.

THE END